

		At point D: object just reached the ground for the second time
4	B	Area under the force-time graph = change in momentum = impulse Total area = $(5 \times 0.5) + (15 \times 0.5) = 10$ $\Delta p = m \Delta v = 10$ $\Delta v = v - 0 \rightarrow v = 10 / 4 = 2.5 \text{ m s}^{-1}$ $2.5 \text{ m s}^{-1} + 2.5 \text{ m s}^{-1} = 5.0 \text{ m s}^{-1}$
5	A	By conservation of momentum